

Data Stream Processing

COMPSCI 752: Big Data Management

University of Auckland

(Credits to Ninh Pham)

real time data.

Slides are collected and edited from <https://vasia.github.io/dspa20/index.html>
and <https://homepages.cwi.nl/~boncz/bads/stream.shtml>

Auckland, May 2025

Outline

- Introduction
 - Definition, applications, challenges
- Data stream processing
 - Data stream modelling
 - Data stream processing

What is data stream?

- Large data volume, likely structured, arriving at a very high rate (high enough that machine cannot keep up it)
- Definition (Golab & Ozsú, 2003):
 - A data stream is a real-time, continuous, ordered (implicitly by arrival time of explicitly by timestamp) sequence of items
 - It is impossible to control the order in which items arrive, nor it is feasible to locally store a stream in its entirety

① 难排序

② 快, 难存

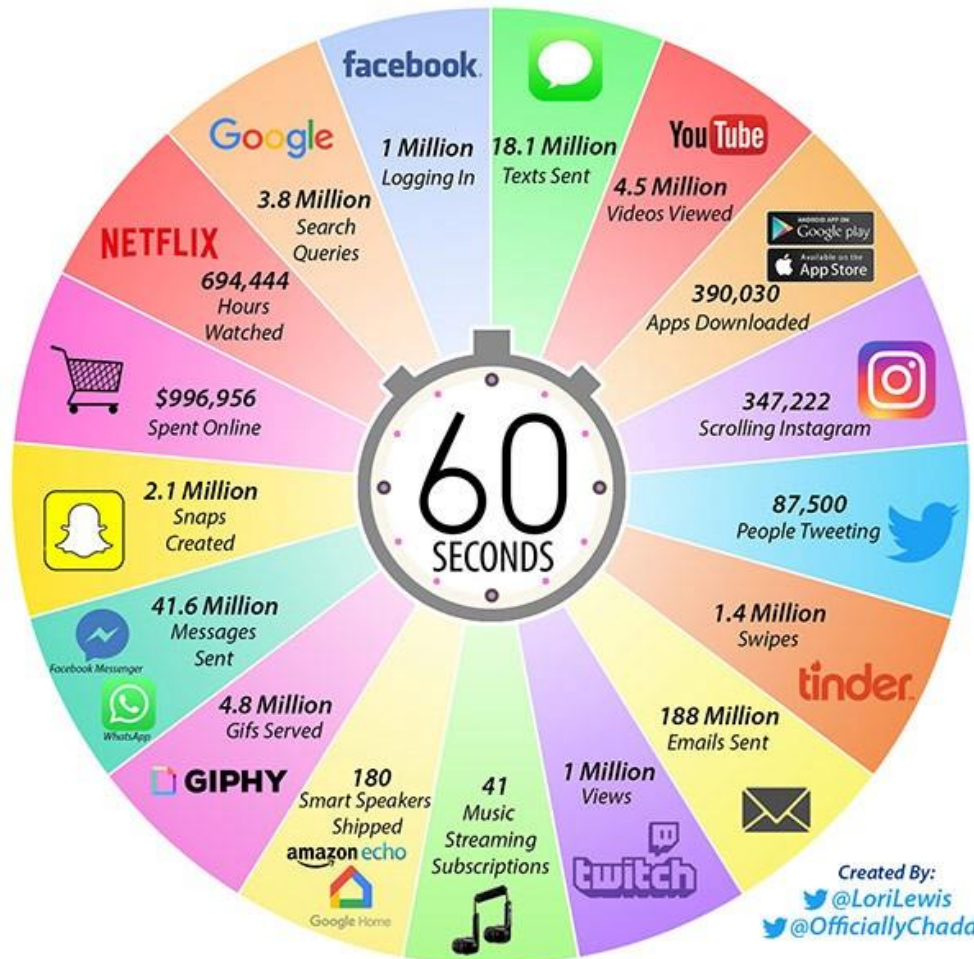
Massive data stream

- We do not know entire data set in advance
- Data arrives in the streaming fashion and if it is not processed immediately, it will be lost forever

THE INTERNET IN **2023** EVERY MINUTE



2019 *This Is What Happens In An Internet Minute*



2021 *This Is What Happens In An Internet Minute*

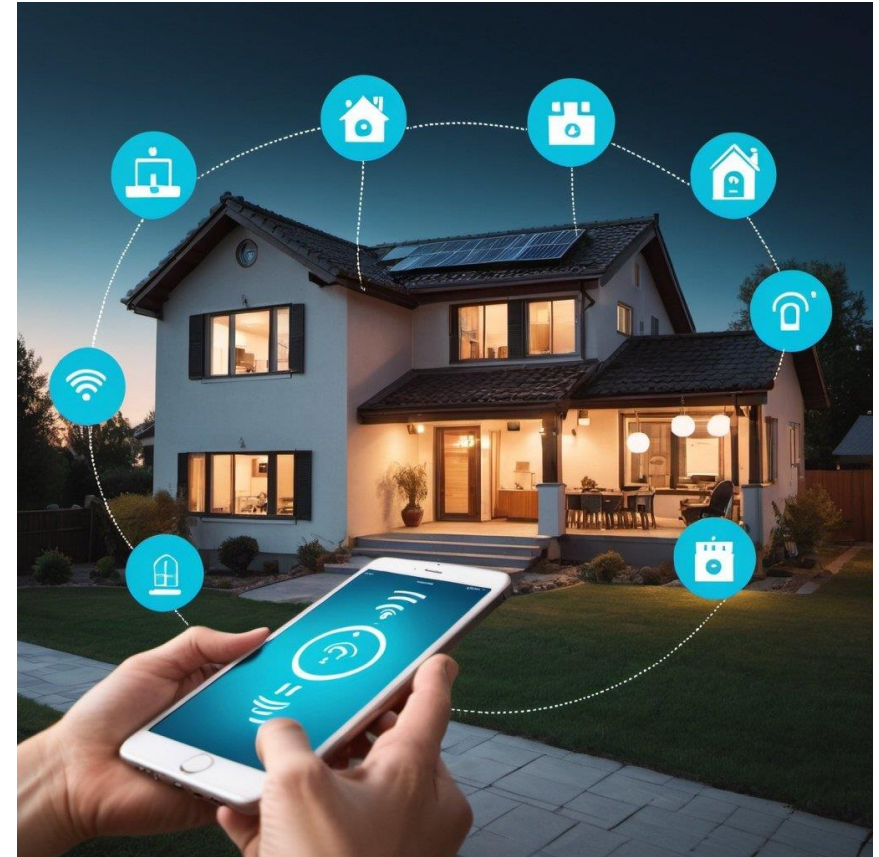


Why do we need data stream?

- Online, real-time processing
- Potential objectives:
 - Event detection and reaction
 - Fast and potentially approximate online aggregation and analytics at different granularities
- Various applications:
 - Network management, telecommunications
 - Load balancing in distributed systems
 - Stock monitoring, finance, fraud detection
 - Online data mining (click stream analysis)

Data stream applications

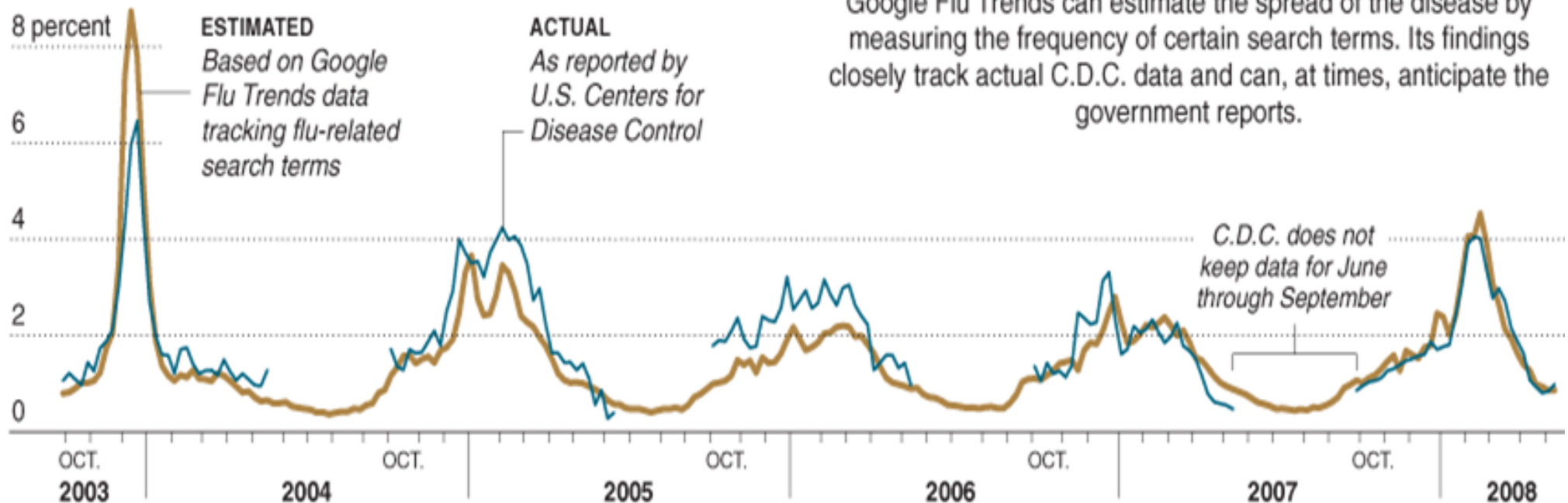
- Sensor networks:
 - Many sensors feeding into a mobile devices
 - What occurs if the mobile device does not have enough computational resources for real-time event detection?



Data stream applications

- Google query streams:
 - What queries are more frequent today than yesterday?
 - Estimate influenza activities by aggregating Google search queries

PERCENT OF HEALTH VISITS FOR FLU-LIKE SYMPTOMS *Mid-Atlantic region*



Using Google to Monitor the Flu

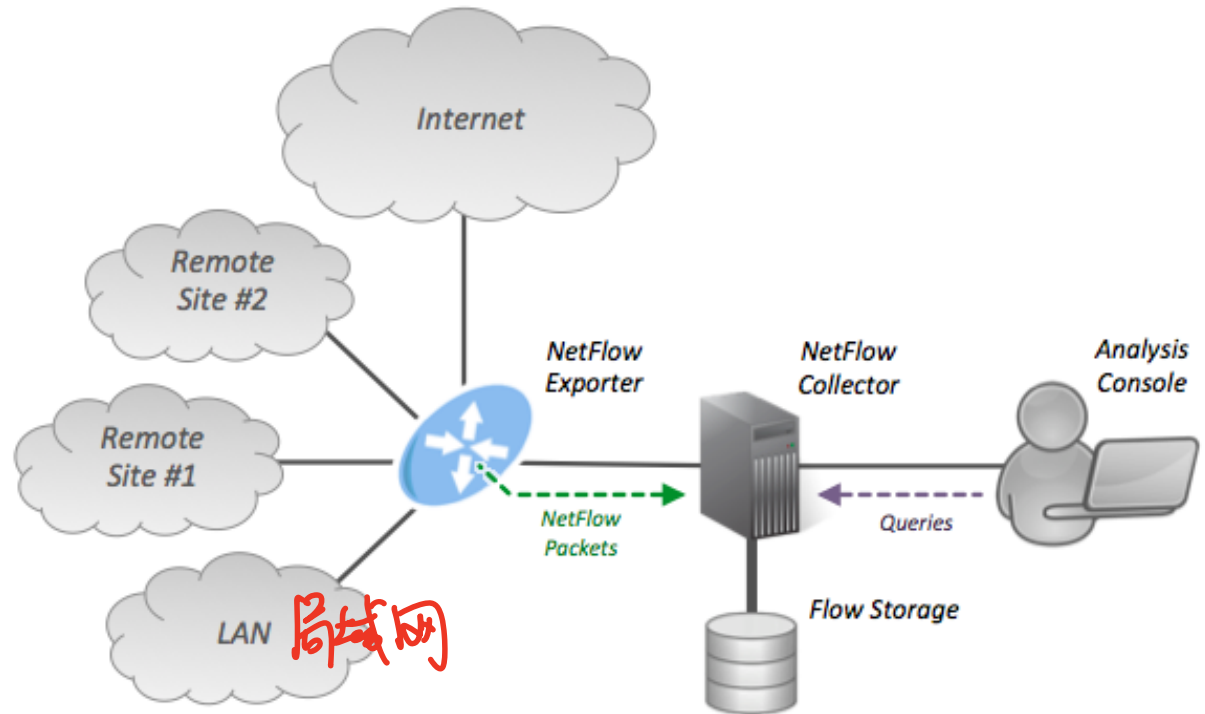
Google Flu Trends can estimate the spread of the disease by measuring the frequency of certain search terms. Its findings closely track actual C.D.C. data and can, at times, anticipate the government reports.

Data stream applications

- Yahoo click streams:
 - What pages are getting an unusual number of hits in the past hour?
- Mining social network:
 - Look for trending topics on Twitter, Facebook based on news feed
- IP packets monitored at a switch in network:
 - Detect denial-of-service attacks
 - Gather information for optimal routing

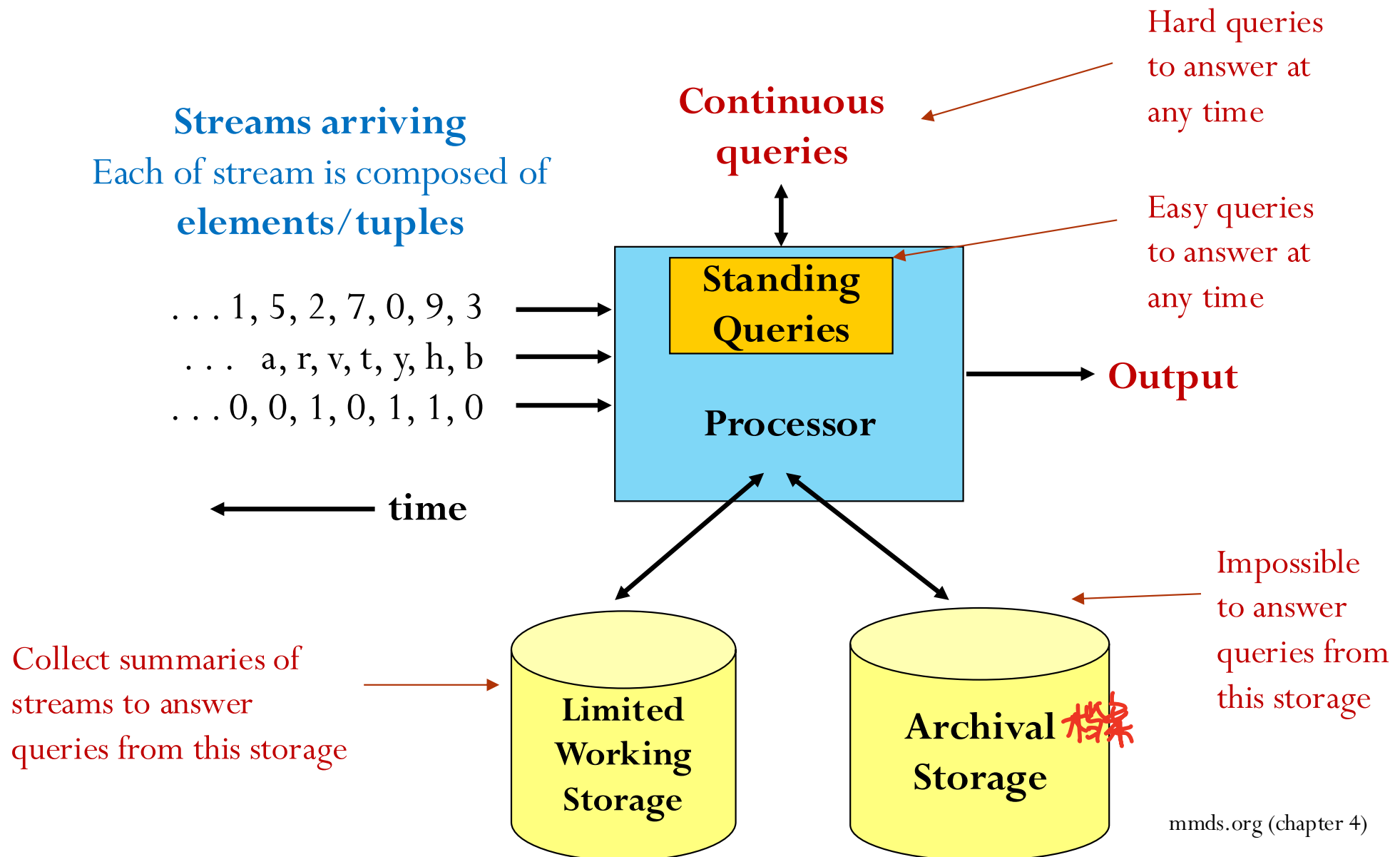
IP network monitoring application

- 24x7 IP packet/flow data streams at network elements
- Massive stream arriving at rapid rates
 - AT&T collects ~ 1 TB of Netflow data per day
 - Off-line analysis is very slow and expensive



Source	Destination	Duration	Bytes	Protocol
10.1.0.2	16.2.3.7	12	20K	http
18.6.7.1	12.4.0.3	16	24K	http
13.9.4.3	11.6.8.2	15	20K	http
15.2.2.9	17.1.2.1	19	40K	http
12.4.3.8	14.8.7.4	26	58K	http
10.5.1.3	13.0.0.1	27	100K	ftp
11.1.0.6	10.3.4.5	32	300K	ftp
19.7.1.2	16.5.5.8	18	80K	ftp

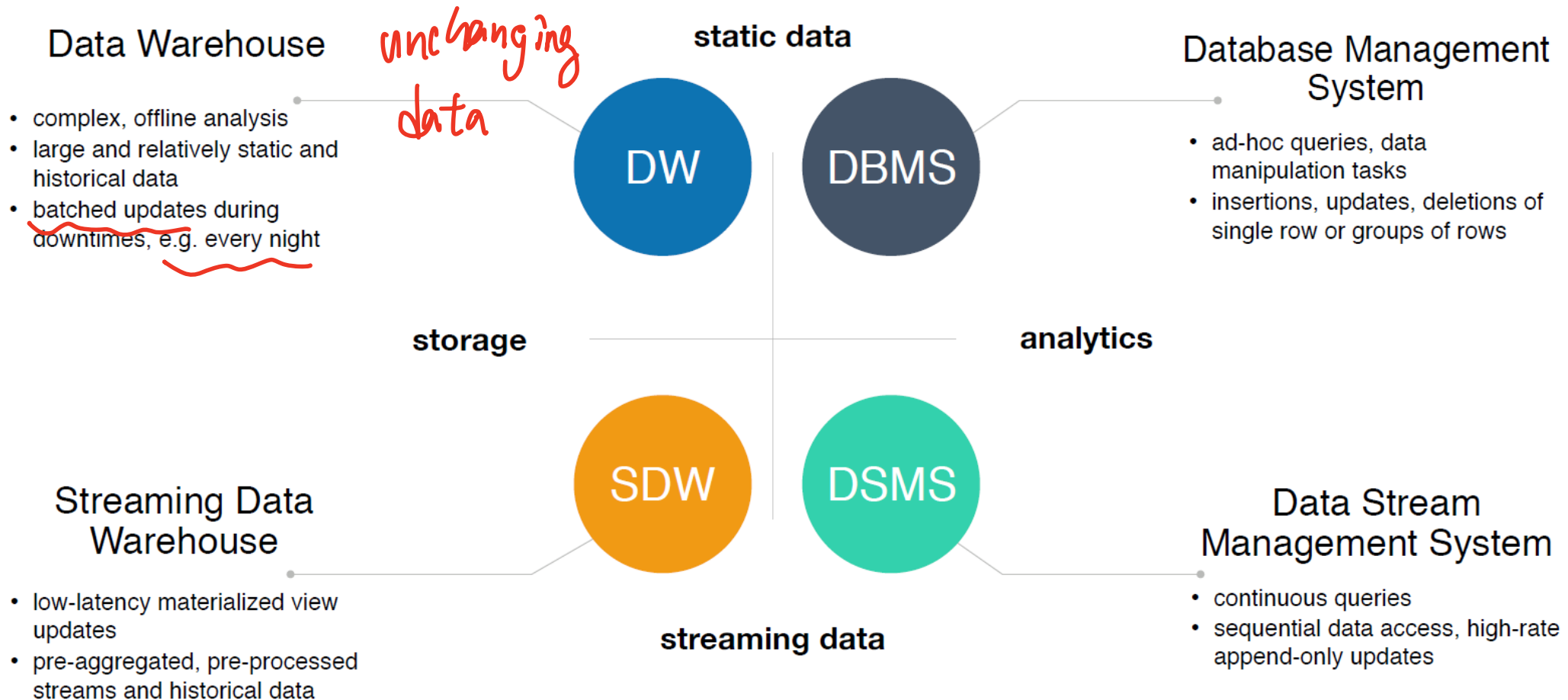
Stream processing model



Structure of data stream

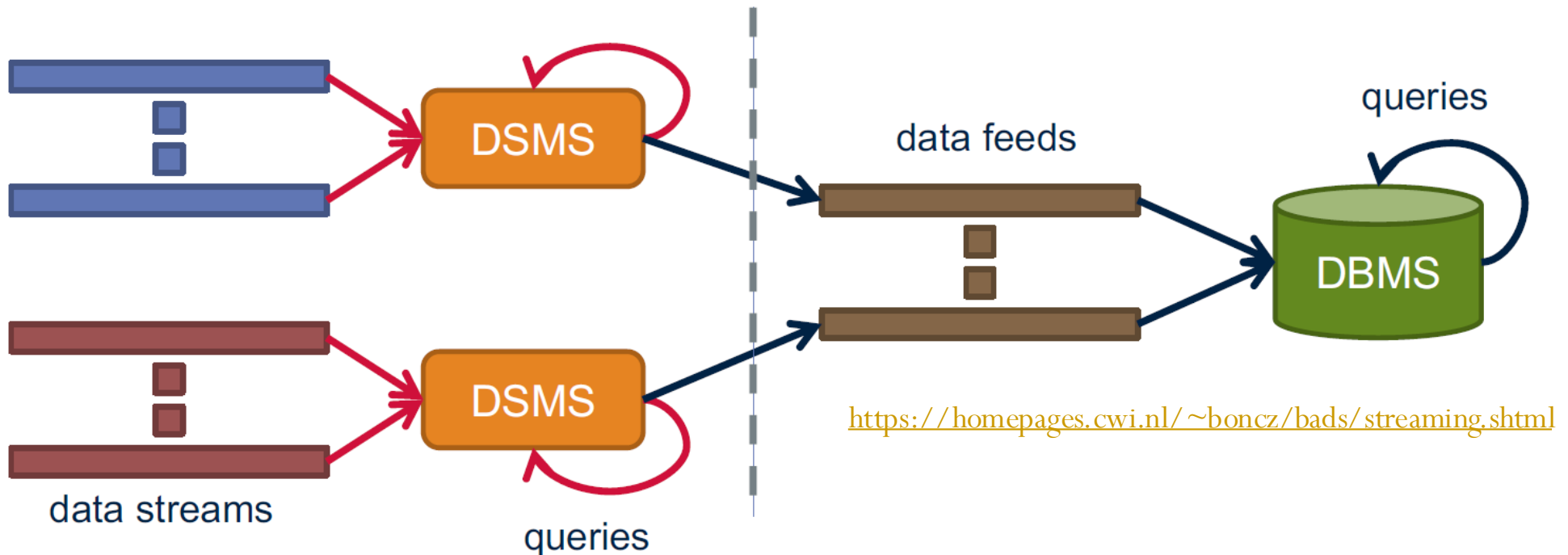
- Infinite ^{无限} sequence of items (elements) which have the same structured information, e.g.
 - Tuple: (Source, Destination, Duration, Bytes, Protocol)
 - Object: a user ID or a search query
- Timestamp:
 - ^{明确的} Explicit (date/time field in data)
 - Implicit (arrival time as a sequence of integers)
_{隐式的}
- Unbounded length, no control of arrival order and data rate

Data management approaches



<https://vasia.github.io/dspa20/index.html>

DSMS vs. DBMS



- Data stream management system (DSMS): Voluminous streams-in, reduced streams-out
- Database management system (DBMS): Output of DSMS can be treated as data feeds to database

DSMS vs. DBMS

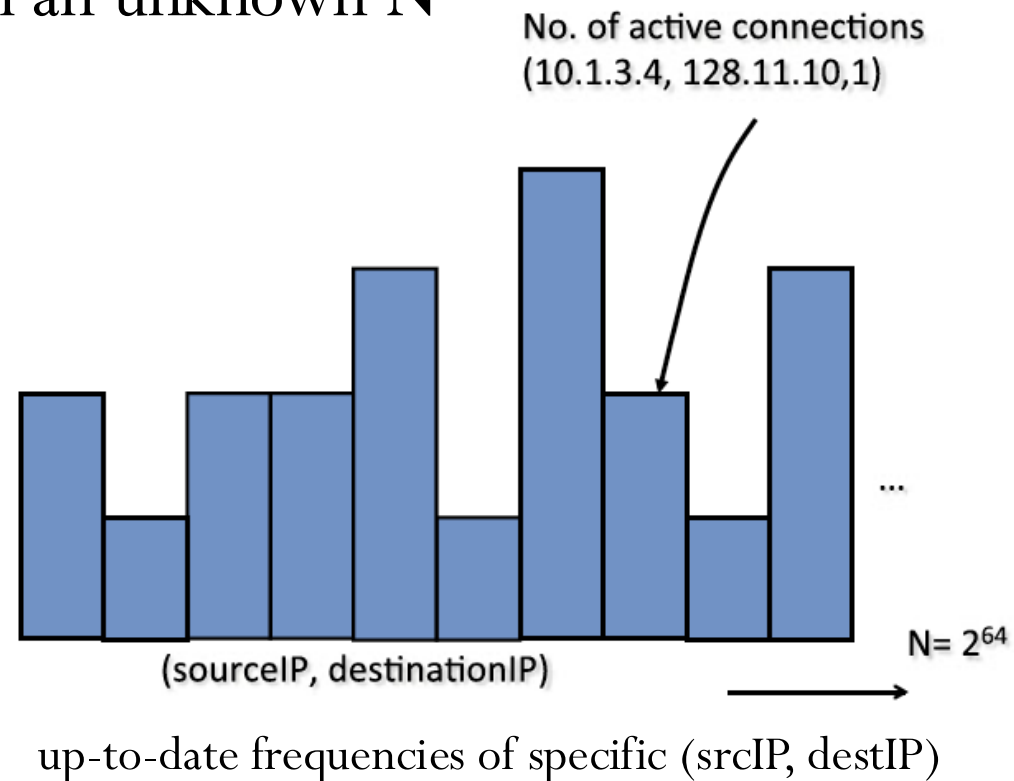
	DBMS	DSMS
Data	persistent relations	streams
Data Access	<u>random</u>	<u>sequential</u> , <u>single-pass</u>
Updates	arbitrary	append-only
Update rates	relatively low	high, bursty
Processing Model	query-driven / pull-based	<u>data-driven</u> / push-based
Queries	ad-hoc 特别指定	continuous
Latency 延迟	relatively high	low

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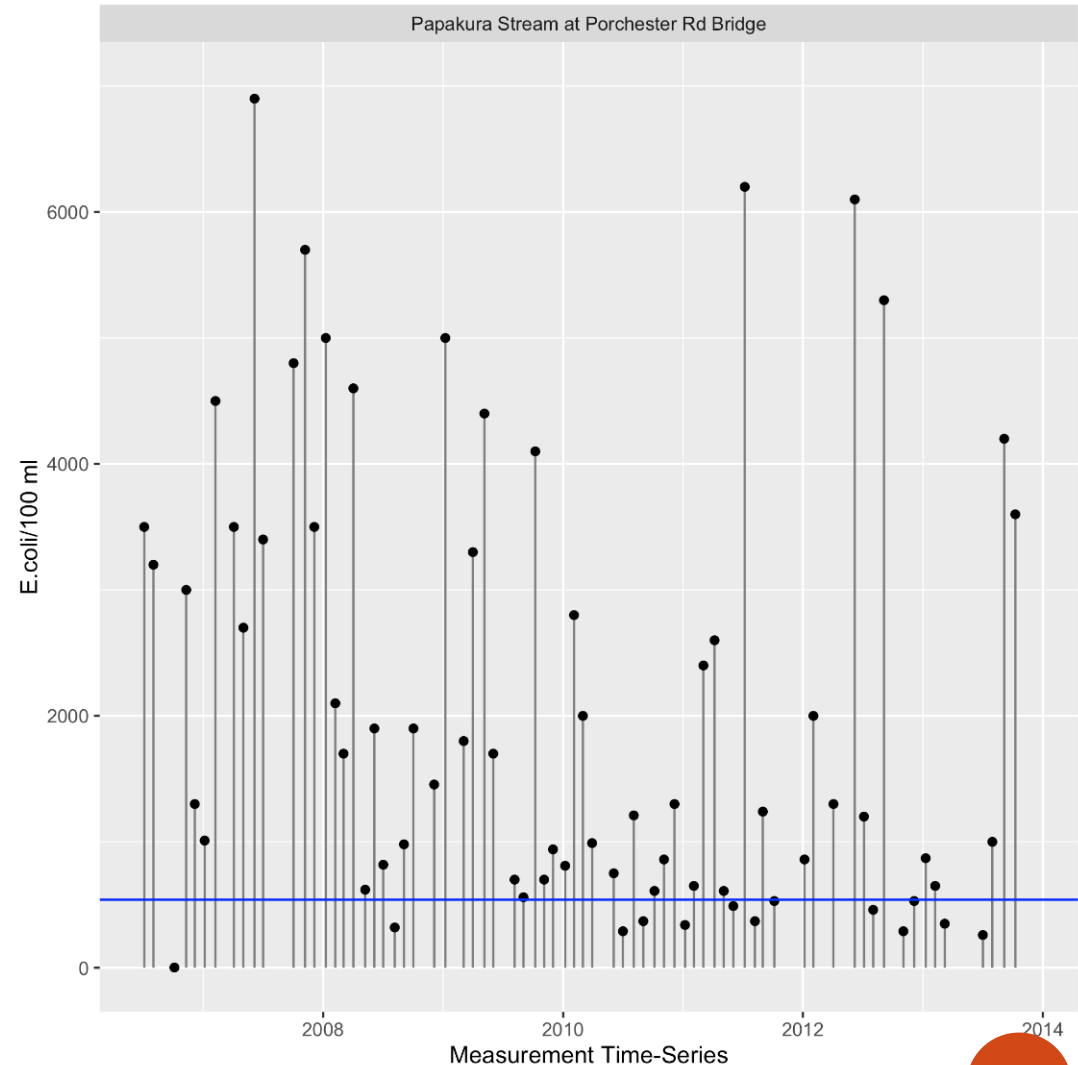
Modeling data stream

- A stream can be viewed as never-ending updates of one-dimensional vector $A[1..N]$ with an unknown N
- The j -th update with the form $(\mathbf{k}, \mathbf{c}[j])$ modifies the $\mathbf{k}$ -th entry of $\mathbf{A}$ as $A[\mathbf{k}] \leftarrow A[\mathbf{k}] + \mathbf{c}[j]$



Time-series model

- The j -th update $(j, A[j])$
 - Updates arriving in increasing order of j as a time series
 - Entries of A are observed with the increasing index
- Drawback:
 - Cannot change the past entries
 - Only practical value for modeling time series data (sensor data, stock data, etc)



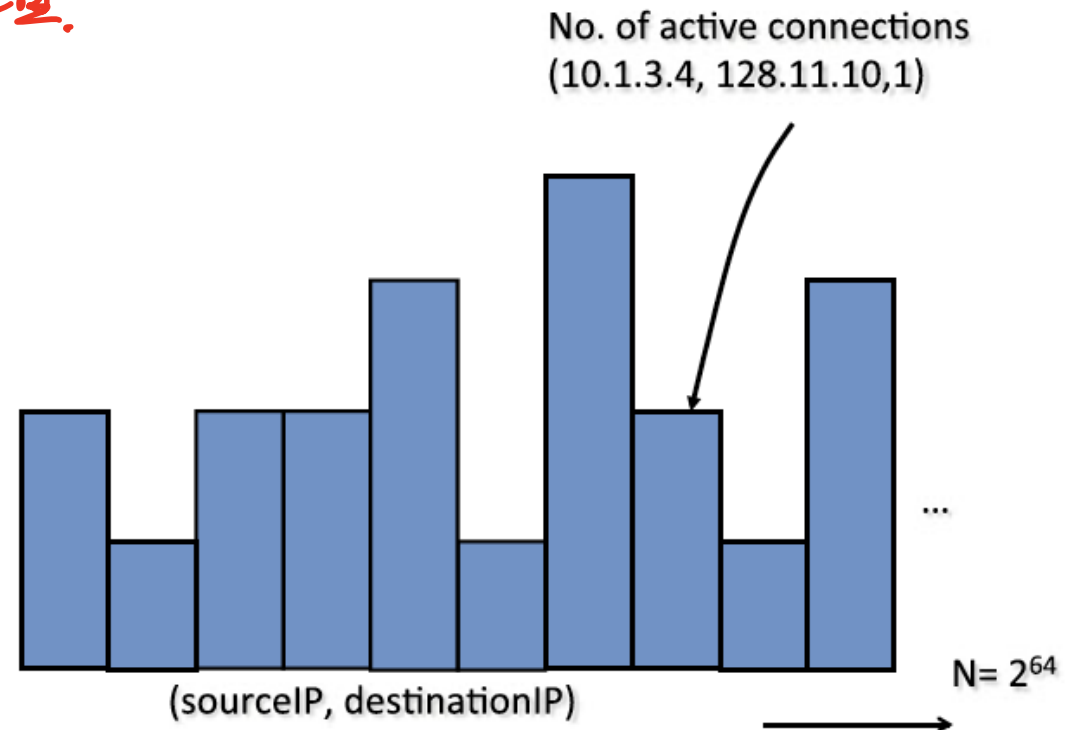
Cash-register model

- Allow multiple updates that only increment an entry in A : 增量.

- The j -th update $(k, c[j])$ increases $A[k]$ by a value $c[j] \geq 0$ 值.

- Drawback:

- Only practical value for insertion-only data stream since the old data are still in the history



<https://vasia.github.io/dspa20/index.html>

只能增加

闸机 Turnstile model

can ↑, ↓

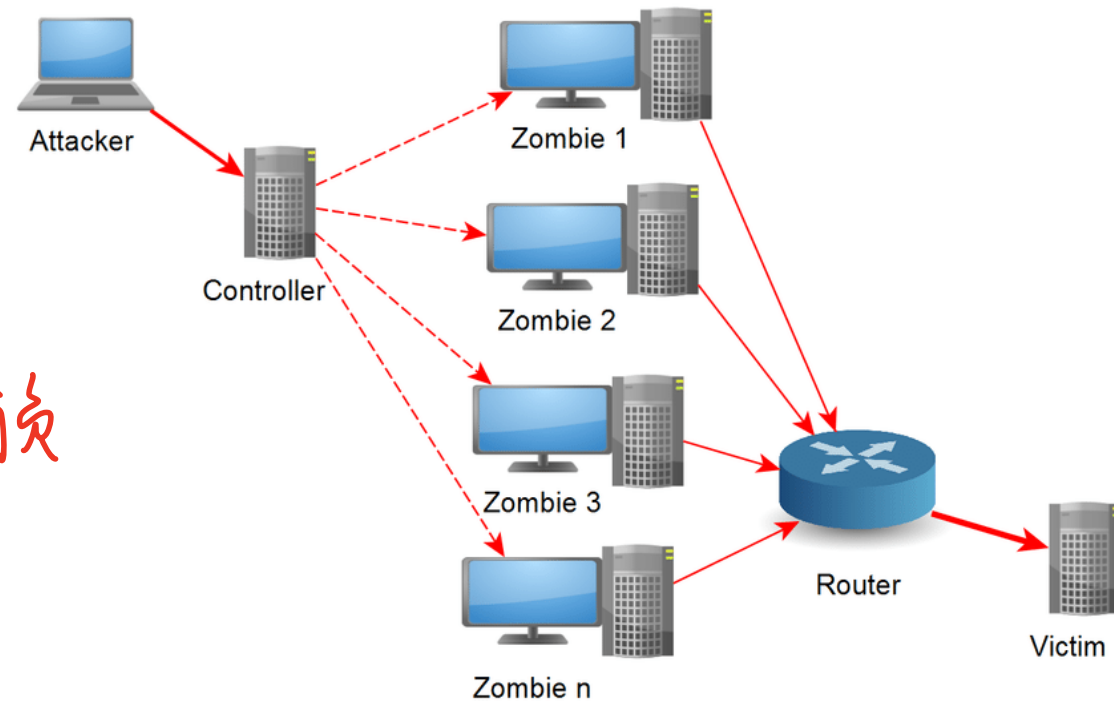
- Allow multiple updates that increment/decrement an entry in A :

- The j -th update $(k, c[j])$ updates $A[k]$ by $A[k] + c[j]$ for any $c[j]$

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- Drawback:

- It is challenging to design space and time-efficient algorithms for this general model

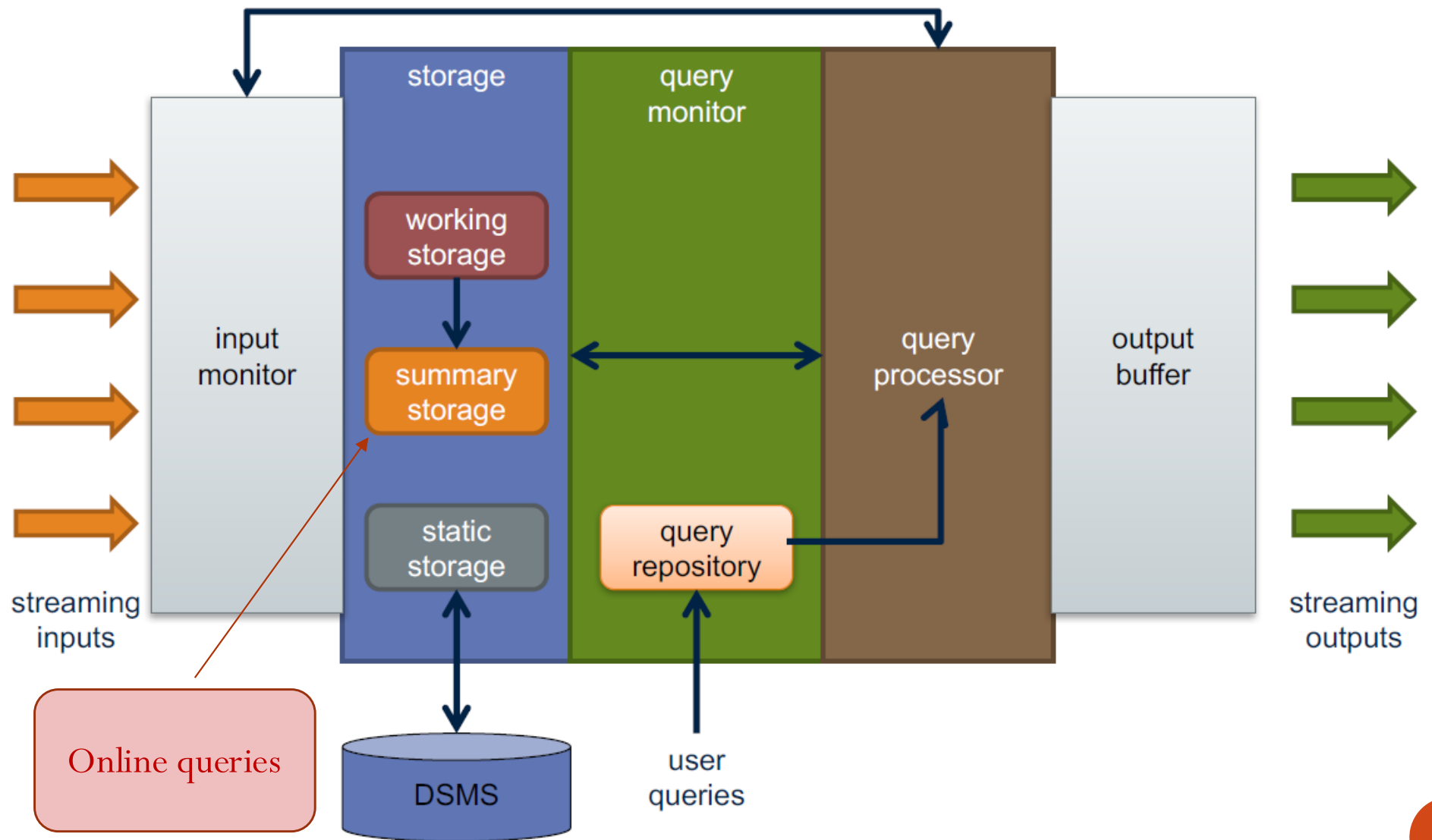


DOS attack, Obinna Igbe et al.'17

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DSMS architecture



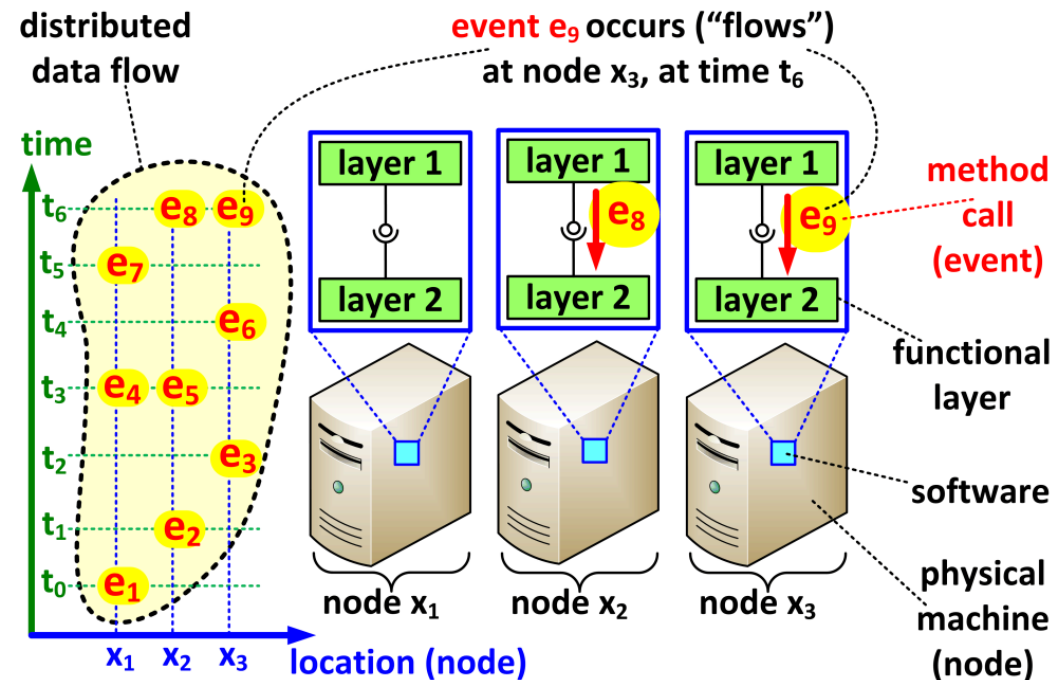
Challenges

- Stream properties:

- Infinite and non-stationary
- Rapid rate, never-ending, so impossible to store the entire stream accessibly

- How do we process the stream using a distributed dataflow to answer queries as quickly as possible?

- Exact solutions are possible by parsing and manipulating inputs on-the-fly
- Distributed dataflow systems: Spark, TensorFlow, Flink, etc



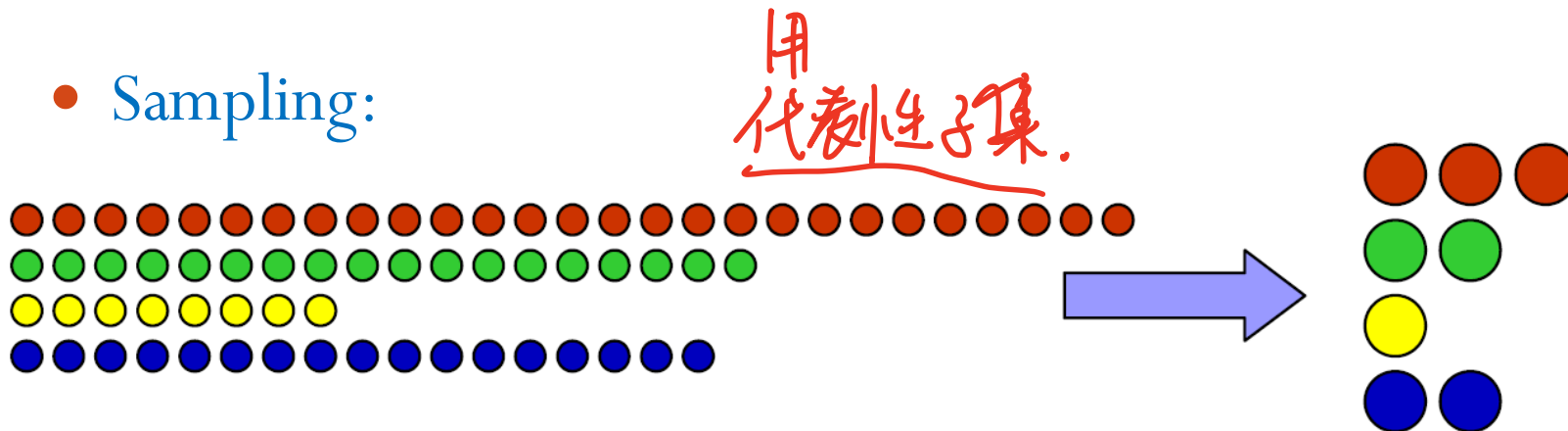
Challenges

- Stream properties:
 - Infinite and non-stationary
 - Rapid rate, never-ending, so impossible to store the entire stream accessibly
- How do we process the stream using a limited amount of memory to answer queries in real-time manner?
 - Exact solutions are often impossible without entirely storing data
 - An approximation answer suffices with theoretical guarantees

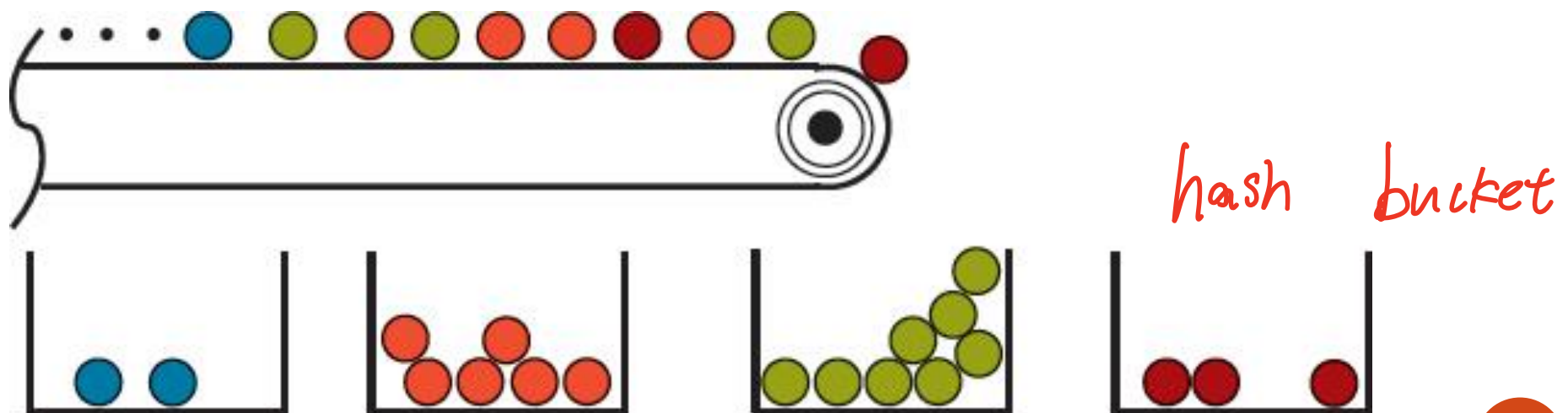


Basic probabilistic algorithms

- Sampling:



- Hashing:



References

- Lukasz Golab and M. Tamer Özsu. **Issues in data stream management**. SIGMOD Rec. 32, 2 (June 2003)
- Michael Stonebraker, Ugur Çetintemel, and Stan Zdonik. **The 8 requirements of real-time stream processing**. SIGMOD Rec. 34, 4 (December 2005)
- Data stream lectures:
 - <https://homepages.cwi.nl/~boncz/bads/stream.shtml>
 - <https://vasia.github.io/dspa20/lectures/dspa20-1.pdf>